CHEMISTRY

PAPER 1 (THEORY)

Maximum Marks: 70 **Time Allowed: Three Hours**

Reading Time: Additional Fifteen Minutes

Instructions to Candidates

1. You are allowed an additional fifteen minutes for only reading the question paper.
2. You must NOT start writing during the reading time.
3. This question paper has 8 printed pages.
4. It is divided into four sections and has twenty one questions in all.
5. Answer all questions.
6. Section A has fourteen subparts. Each question carries 1 mark.
7. While attempting Multiple Choice Questions in Section A, you are required to write only ONE option as the answer.
8. Section B has ten questions. Each question carries 2 marks.
9. Section C has seven questions. Each question carries 3 marks.
10. Section D has three questions. Each question carries 5 marks.
11. Internal choices have been provided in two questions each in Sections B, C and D.
12. The intended marks for questions are given in brackets [ ].
13. All working, including rough work, should be done on the same sheet as, and adjacent to the rest of the answer.
14. Balanced equations must be given wherever possible and diagrams where they are helpful.
15. When solving numerical problems, all essential workings must be shown.
16. In working out problems, use the following data:

    Gas constant R = 1·987 cal deg^{-1} mol^{-1} = 8·314 J K^{-1} mol^{-1} = 0·0821 dm^3 atm K^{-1} mol^{-1}, 1 l atm = 1 dm^3 atm = 101·3 J.

    1 faraday = 96500 coulombs, Avogadro's number = $6·023 \times 10^{23}$.

Instruction to Supervising Examiner

Kindly read aloud the Instructions given above to all the candidates present in the examination hall.

SECTION A – 14 MARKS

Question 1

(A) Fill in the blanks by choosing the appropriate word(s) from those given in the brackets.

[propanone, ethanal, iodoform, chloroform, specific, molar, equivalent, completely filled, partly filled, diamagnetic, paramagnetic, Lucas, Tollens', tertiary, primary, secondary]

(i) Calcium acetate on dry distillation gives _____, which gives _____ on heating with iodine and sodium hydroxide solution. *(Recall)* [1]

(ii) As the concentration of an electrolytic solution is increased, its _____ conductance increases but its _____ conductance decreases. *(Understand)* [1]

(iii) Transition metal ions having _____ d-orbitals are colourless and _____ in nature. [1]
(Understand)

(iv) A mixture of conc. HCl and anhydrous $ZnCl_2$ is called _____ reagent; it gives turbidity immediately with a _____ alcohol. *(Recall)* [1]

(B) Select and write the correct alternative from the choices given below.

(i) If two liquids A and B form a minimum boiling azeotrope at a specific composition, then: [1]
(Understand)

(a) A–B interactions are stronger than A–A and B–B interactions.

(b) A–B interactions are weaker than A–A and B–B interactions.

(c) The mixture shows a negative deviation from Raoult's law.

(d) The vapour pressure of the mixture is lower than that of either pure liquid.

(ii) Which one of the following monosaccharides is a ketose? *(Recall)* [1]

(a) Glucose

(b) Galactose

(c) Fructose

(d) Ribose

(iii) The number of unpaired electrons and the shape of the complex ion $[NiCl_4]^{2-}$ are respectively (atomic number of Ni = 28): *(Apply)* [1]

(a) 2 and tetrahedral

(b) 0 and square planar

(c) 2 and square planar

(d) 1 and tetrahedral

(iv) The rate of a reaction is given by rate = $k[A]^2[B]$. If the concentration of A is increased three times and that of B is increased four times, the rate of the reaction increases by a factor of: [1]
(Apply)

(a) 12

(b) 24

(c) 36

(d) 72

(v) An organic compound [A], C_2H_7N, on treatment with nitrous acid gives an alcohol and nitrogen gas. [A] on warming with chloroform and caustic potash gives [C], which on reduction gives ethylmethylamine. The structure of [A] is: *(Analyse)* [1]

(a) $(CH_3)_2NH$

(b) $(CH_3)_3N$

(c) CH_3NH_2

(d) $CH_3CH_2NH_2$

(vi) Given below are two statements marked Assertion and Reason. Read the two statements carefully and select the correct option. [1]

Assertion: *p*-Nitrophenol is more acidic than phenol.

Reason: The nitro group helps in the stabilisation of the phenoxide ion by dispersal of the negative charge through resonance.

(Analyse)

(a) Both Assertion and Reason are true, and Reason is the correct explanation for Assertion.

(b) Both Assertion and Reason are true, but Reason is not the correct explanation for Assertion.

(c) Assertion is true and Reason is false.

(d) Both Assertion and Reason are false.

(vii) Given below are two statements marked Assertion and Reason. Read the two statements carefully and select the correct option. [1]

Assertion: The molar conductivity of acetic acid rises very sharply on dilution, whereas that of KCl rises only gradually.

Reason: The degree of dissociation of a weak electrolyte increases on dilution.

(Analyse)

(a) Both Assertion and Reason are true, and Reason is the correct explanation for Assertion.

(b) Both Assertion and Reason are true, but Reason is not the correct explanation for Assertion.

(c) Assertion is true and Reason is false.

(d) Both Assertion and Reason are false.

(C) Read the passage carefully and answer the questions that follow.

Phenol is a hydroxy derivative of benzene. In low concentration (0·2 %) it is an antiseptic, whereas a 2 % solution is used as a disinfectant; it is also the starting material for dyes, plastics, drugs and explosives. Because the –OH group activates the benzene ring, phenol undergoes electrophilic substitution easily; its acidity (pK_a about 10) is much greater than that of ethanol (about 16), and the phenoxide ion is a good nucleophile. Phenol can be made on the large scale from chlorobenzene or from cumene, and it is converted into many useful compounds, among them azo dyes and salicylic acid, the raw material of aspirin.

(i) What do you observe when phenol is treated with an alkaline solution of benzenediazonium chloride at 273–278 K? Give the balanced chemical equation. *(Apply)* [1]

(ii) Give the balanced chemical equations for the conversion of phenol into salicylic acid. *(Recall)* [1]

(iii) Phenol is heated with excess conc. H_2SO_4 to form Y. Y reacts with conc. HNO_3 to form Z. Identify Y and Z. *(Analyse)* [1]

SECTION B – 20 MARKS

Question 2

Sourav, a student of Class XII, prepared a 0·01 M H_2SO_4 solution and reported that its conductivity is $2{\cdot}2 \times 10^{-4}$ S cm^{-1} and its equivalent conductivity is 22 S cm^2 eq^{-1}. Are these data justified? Explain. *(Evaluate)* [2]

Question 3

An aromatic compound [A], C_8H_8O, gives a positive 2,4-DNP test and gives a yellow precipitate [B] with iodine and sodium hydroxide solution. [A] does not give Tollens' or Fehling's test. On drastic oxidation with $KMnO_4$, [A] forms a carboxylic acid [C], $C_7H_6O_2$, which is also formed along with [B] in the iodine reaction. When [B] is heated with silver powder, a compound [D] is formed which decolourises bromine in CCl_4. Identify [A], [B], [C] and [D]. *(Analyse)* [2]

Question 4

(i) During a chemistry class, the teacher wrote $[Fe(CN)_6]^{3-}$ on the board and asked the students for the hybridisation and magnetic character of the complex ion (atomic number of Fe = 26). Asha wrote sp^3d^2 and diamagnetic, whereas Ravi wrote d^2sp^3 and paramagnetic. Evaluate Asha's and Ravi's responses. *(Evaluate)* [2]

OR

(ii) Using crystal field theory, answer the following for Mn^{3+} in the octahedral complexes $[Mn(CN)_6]^{3-}$ and $[MnCl_6]^{3-}$ (atomic number of Mn = 25).

(a) Write the electronic configuration of Mn^{3+} in each complex in terms of t_{2g} and e_g. *(Understand)* [1]

(b) Calculate the crystal field stabilisation energy of each complex in terms of Δ_o. (Ignore the pairing energy.) *(Apply)* [1]

Question 5

Iodine-131 is used in the treatment of thyroid disorders. What fraction of a given amount of iodine-131 remains after 40 days if its half-life is 8 days? Assume that none of it is eliminated by the body. *(Apply)* [2]

Question 6

Answer the following:

(i) How can ethylamine be converted into methylamine? (Write the chemical equations.) *(Apply)* [1]

(ii) Haloalkanes react with KCN to give alkyl cyanides as the main product, but with AgCN they give isocyanides. Give a reason. *(Understand)* [1]

Question 7

Give reasons for the following:

(i) The separation of zirconium (Z = 40) and hafnium (Z = 72) is difficult. *(Understand)* [1]

(ii) Copper is considered a transition element but zinc is not. (Atomic number of Cu = 29, Zn = 30) *(Analyse)* [1]

Question 8

Write the Nernst equation and calculate the emf of the following cell at 298 K: [2]

$$Cd(s) \mid Cd^{2+}(1 \times 10^{-2}\ M) \parallel Cu^{2+}(1 \times 10^{-4}\ M) \mid Cu(s)$$

Given: $E°(Cd^{2+}/Cd) = -0{\cdot}403$ V, $E°(Cu^{2+}/Cu) = +0{\cdot}337$ V *(Apply)*

Question 9

(i) Illustrate the following reactions with an example in each case:

(a) Coupling reaction *(Recall)* [1]

(b) Balz–Schiemann reaction *(Recall)* [1]

OR

(ii) An aromatic carboxylic acid [A] which readily sublimes on heating gives, on heating with aqueous ammonia, a compound [B]. [B] on heating with Br_2 and KOH forms a compound [C], C_6H_7N. Write the structures of [A], [B] and [C] and name the reaction by which [B] is converted into [C]. *(Understand)* [2]

Question 10

A hydrogenation reaction proceeds at a certain rate at 600 K. In the presence of a catalyst the same rate is obtained at 450 K. The catalyst lowers the activation energy by 30 kJ mol^{-1}. Calculate the activation energy of the uncatalysed reaction. (Assume that the pre-exponential factor is the same in both cases.) *(Apply)* [2]

Question 11

A solution of 0·920 g of a compound in 20·0 g of benzene freezes 1·28 K below the freezing point of pure benzene. (K_f for benzene = 5·12 K kg mol^{-1})

(i) Calculate the experimental molecular mass of the compound. *(Apply)* [1]

(ii) If the theoretical molecular mass of the compound is 92·0 g mol^{-1}, state whether it undergoes association or dissociation in benzene. *(Analyse)* [1]

SECTION C – 21 MARKS

Question 12

Write the chemical equations for the following reactions and name the products:

(i) Chlorobenzene is heated with conc. HNO_3 in the presence of conc. H_2SO_4. *(Apply)* [1]

(ii) Propan-1-ol is treated with thionyl chloride in the presence of pyridine. *(Apply)* [1]

(iii) Chloroform is slowly oxidised by air in the presence of sunlight. *(Apply)* [1]

Question 13

The following data were obtained for the reaction 2P + Q → products at 298 K:

Experiment	[P] (mol L^{-1})	[Q] (mol L^{-1})	Rate (mol L^{-1} min^{-1})
1	0·10	0·05	$3·0 \times 10^{-3}$
2	0·20	0·05	$6·0 \times 10^{-3}$
3	0·40	0·10	$4·8 \times 10^{-2}$

(i) What is the order of the reaction with respect to P and with respect to Q? *(Analyse)* [1]

(ii) Write the rate law and state the overall order. *(Understand)* [1]

(iii) Calculate the rate constant. *(Apply)* [1]

Question 14

(i) Write a chemical test to distinguish between the following pairs of compounds:

(a) Phenol and cyclohexanol *(Apply)* [1]

(b) Methanol and ethanol *(Apply)* [1]

(c) Ethanol and propan-2-ol *(Apply)* [1]

OR

(ii) How will you convert the following? (Write chemical equations.)

(a) Ethyl bromide to diethyl ether *(Recall)* [1]

(b) Chlorobenzene to phenol *(Recall)* [1]

(c) Ethanal to propan-2-ol *(Recall)* [1]

Question 15

(i) Complete and balance: $K_2Cr_2O_7 + H_2SO_4 + KI \rightarrow$ ____ + ____ + ____ + ____ *(Understand)* [1]

(ii) Complete and balance: $KMnO_4 + H_2SO_4 + FeSO_4 \rightarrow$ ____ + ____ + ____ + ____ *(Understand)* [1]

(iii) Explain how the colour of a potassium dichromate solution depends on the pH of the solution. *(Analyse)* [1]

Question 16

(i) Give one example each of a fibrous protein and a globular protein. *(Recall)* [1]

(ii) Draw the peptide linkage between the two amino acids alanine, $H_2N–CH(CH_3)–COOH$, and glycine, $H_2N–CH_2–COOH$, in the dipeptide alanylglycine. *(Understand)* [1]

(iii) Justify, with the help of a chemical reaction, that glucose is an aldose. *(Understand)* [1]

Question 17

(i) Answer the following:

(a) What type of cell is a lead storage battery? Name its electrolyte. *(Recall)* [1]

(b) Write the anode and cathode reactions occurring during the use (discharge) of the lead storage battery. *(Recall)* [1]

(c) Two metallic elements A and B have oxidation potentials +0·76 V and −0·34 V respectively. What would you expect if a piece of element A is added to an aqueous solution of a salt of element B? *(Analyse)* [1]

OR

(ii) The conductivity of a 0·0050 M solution of acetic acid at 298 K is $1·1 \times 10^{-4}$ S cm^{-1}. The molar conductivity of acetic acid at infinite dilution is 390·5 S cm^2 mol^{-1}. Calculate:

(a) the molar conductivity of the solution *(Apply)* [1]

(b) the degree of dissociation *(Apply)* [1]

(c) its dissociation constant *(Apply)* [1]

Question 18

Write the chemical equations for the following conversions: *(Recall)*

(i) Nitrobenzene to phenol [1]

(ii) Aniline to fluorobenzene [1]

(iii) Methyl bromide to ethylamine [1]

SECTION D – 15 MARKS

Question 19

(i) Give balanced chemical equations, with conditions where necessary, for the following name reactions:

(a) Wolff–Kishner reduction *(Recall)* [1]

(b) Perkin reaction *(Recall)* [1]

(c) Etard reaction *(Recall)* [1]

(ii) A ketone [A], C_3H_6O, undergoes the haloform reaction and on reduction gives [B]. [B] on heating with conc. H_2SO_4 at 443 K gives [C], which on ozonolysis (O_3, then Zn/H_2O) gives two products, [D] and [E]. Identify [A], [B], [C], [D] and [E]. *(Analyse)* [2]

OR

(iii) Give balanced chemical equations for the following reactions:

(a) Benzoin condensation *(Recall)* [1]

(b) Cannizzaro reaction of formaldehyde *(Recall)* [1]

(c) Clemmensen reduction of acetophenone *(Recall)* [1]

(iv) Identify [A] and [B] in each of the following: [2]
(a) $C_6H_5CH_2CH_3 \rightarrow$ (alk. $KMnO_4$, then H^+) [A] $\rightarrow$ (NaOH/CaO, Δ) [B]
(b) $CH_3COOH \rightarrow$ ($SOCl_2$) [A] $\rightarrow$ (H_2, Pd–$BaSO_4$) [B] *(Understand)*

Question 20

(i) Write the IUPAC name of (a) $[Pt(NH_3)_2Cl(NO_2)]$ and (b) $Fe_4[Fe(CN)_6]_3$. *(Recall)* [1]

(ii) State the type of isomerism shown by $[Co(NH_3)_5Br]SO_4$ and $[Co(NH_3)_5(SO_4)]Br$. How can the two be distinguished by a chemical test? *(Analyse)* [1]

(iii) Using valence bond theory, predict the hybridisation and the magnetic behaviour of (a) $[Co(NH_3)_6]^{3+}$ and (b) $[CoF_6]^{3-}$. (Atomic number of Co = 27) *(Apply)* [1]

(iv) Draw the optical isomers of $[CoCl_2(en)_2]^+$. *(Understand)* [2]

Question 21

(i) Answer the following:

(a) The vapour pressures of pure liquids A and B are 500 and 800 mm Hg respectively at 350 K. Find the composition of the liquid mixture if the total vapour pressure is 680 mm Hg, and the composition of the vapour phase. *(Apply)* [2]

(b) The osmotic pressure of a solution of 40 g of a polymer in 500 mL of solution varies linearly with temperature, the slope of the plot of π (atm) against T (K) being $2{\cdot}05 \times 10^{-4}$ atm K^{-1}. (R = 0·0821 L atm K^{-1} mol^{-1}) (1) Calculate the molar mass of the polymer. (2) By what factor does the osmotic pressure change if the temperature (in K) is doubled? *(Apply)* [2]

(c) Three samples of fruit juice A, B and C contain glucose as the only sugar, at concentrations 0·15 M, 0·40 M and 0·25 M respectively. The freezing point will be highest for: (a) A (b) B (c) C (d) all have the same freezing point. *(Apply)* [1]

OR

(ii) Answer the following:

(a) The Henry's law constants (K_H) of He, O_2 and N_2 in water are $1{\cdot}514 \times 10^5$, $5{\cdot}32 \times 10^4$ and $2{\cdot}66 \times 10^5$ atm respectively. Arrange the gases in the increasing order of their solubility. *(Apply)* [1]

(b) At what partial pressure of O_2 would its mole fraction in water be $4{\cdot}0 \times 10^{-6}$? *(Apply)* [1]

(c) Ramesh dissolved 10 g of glucose in 500 mL of water and observed a boiling point of 100·06 °C. He then prepared two more solutions: (1) 10 g of sucrose (molar mass 342 g mol^{-1}) in 500 mL of water and (2) 10 g of glucose in 500 mL of acetic acid. Will the boiling point be the same in each case? Justify your answer. *(Evaluate)* [2]

(d) Which colligative property is the most suitable for determining the molar mass of a polymer? Justify your answer. *(Understand)* [1]

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